Governing by Artifacts: An Auditable Constraint-Based Framework for Contestable Educational Decision Systems

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ABSTRACT

Automated educational decision support requires evidence that decisions followed an explicit institutional policy. We present Compass-ECL-MAS, an artifact-oriented framework combining empirical constraints, policy versioning, deterministic selection over feasible Pareto candidates, and replay of Top-K decisions with abstention. We evaluate governance mechanics using fixed scored cases from ASSISTments, OULAD and a synthetic stress test. The implemented tasks predict correctness or student success; they do not evaluate early-warning intervention targeting. Reclassifying unselected cases changes automation coverage without necessarily changing selection utility, exposing an institutional choice that must be audited. We reconcile historical outputs with a revised implementation that specifies entropy in bits, stable tie rules, artifact verification and an authorization gate for infeasible candidates. An exploratory, counterbalanced study ($N = 16$; eight standardized cases) shows diagnostic and remediation accuracy of 0.5625 under logs/cards and 0.921875 under structured evidence; correctness on these two endpoints coincides in every response. The study evaluates the evidence bundle as a whole, not an independently isolated replay effect or field recourse. We release reproducible analyses and explicitly distinguish fixed-score replay, new inference, and unsupported feature-dependent exceptions.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → **Computing / technology policy**; *Government technology policy.*

KEYWORDS

contestability, algorithmic accountability, procedural justice, public sector AI, educational decision support, fairness, audit trails, policy versioning, abstention

1 NOTATION AND CORE DEFINITIONS

Table 1 summarizes the symbols used throughout the paper. We emphasize that all fairness metrics are defined *under the deployed decision rule* (Top-K with abstention), not merely on raw scores.

2 INTRODUCTION

Why FAccT should care. Public education agencies increasingly deploy (or procure) predictive decision support to triage scarce interventions: tutoring slots, counseling capacity, special-needs assessments, dropout prevention programs, and scholarship support. When these systems fail, the harm is not only predictive error—it is a *crisis of legitimacy*: affected students and families cannot understand, contest, or obtain meaningful remedy.

Table 1: Notation summary.

Symbol	Meaning
n	number of instances in a dataset split
x_i, y_i, a_i	features, label, sensitive attribute for instance i
f_θ	predictive model with parameters θ
s_i	score $s_i = f_\theta(x_i) \in [0, 1]$
u_i	predictive entropy in bits, $0 \le u_i \le 1$
γ	abstention threshold (eligibility)
$\mathcal{E}_\gamma$	eligible set $\{i : u_i \le \gamma\}$
K	selection capacity ceiling (Top-K)
TopK$(\cdot)$	selected intervention set under Top-K
C	automation coverage $(n_{\text{sel}} + n_{\text{rej}})/n$
$\Delta_{\text{EOpp@K}}$	equality of opportunity gap under Top-K
$\Delta_{\text{DP@K}}$	demographic parity gap under Top-K
H	explanation entropy proxy (global)
d_{eff}	effective feature count proxy

Recent episodes illustrate this legitimacy gap. In 2020, the UK A-level "grading algorithm" was rolled out with limited contestation capacity and generated large-scale, socially salient downgrades that triggered public backlash and a policy U-turn [23]. In U.S. K–12 settings, early-warning models have drawn scrutiny for inequitable risk assignments and opaque governance (e.g., Wisconsin's dropout early-warning system) [9]. And in adjacent public-sector risk profiling, courts have struck down systems for insufficient transparency and accountability (e.g., the Dutch SyRI ruling) [7]. These cases are not outliers: they expose a recurring pattern where *automation arrives before contestability and accountability.*

From metrics to contestability. Responsible-AI frameworks and toolkits have matured rapidly, but many deployments still reduce governance to model-level metrics and checklists. This creates a risk of *fairwashing*: institutions claim ethical compliance via selective indicators while offering weak recourse, unclear responsibility for trade-offs, and poor auditability [5, 13, 17, 22]. In public education, where labels and historical measurements can embed structural inequities [3], governance must be treated as an institutional procedure, not a one-off model tuning exercise.

Our approach: governing by artifacts. We propose Compass-ECL-MAS, a governance-first layer that turns ethical requirements into *versioned policy* and produces *auditable artifacts* supporting oversight, appeals, and iterative revision. The central design premise is simple: *if a decision can be contested, the system*

must ship the artifacts required to contest it. Concretely, the framework (i) encodes multi-dimensional requirements as computable constraints, (ii) implements *abstention as a procedural safeguard* for uncertain cases (routing to human review), and (iii) selects deployable configurations via an explicit, institution-defined policy over Pareto-feasible candidates.

2.1 Contributions

Scope. We make *mechanism-level* claims about governance infrastructure: candidate sweeps, constraint checking, deterministic selection, artifact emission, and replayability for operational contestability. We do *not* claim downstream social impact (e.g., improved educational outcomes, reduced inequity in deployment, or field adoption). Our artifacts are *necessary but not sufficient* for substantive recourse: institutions still must provide accessible processes, capacity, and legal safeguards.

This paper contributes *auditable infrastructure* for contestable public educational decision support:

- **Abstention as procedural safeguard:** We formalize a Top-K decision rule with uncertainty-triggered abstention that creates a concrete pathway to human review for uncertain or policy-sensitive cases.
- **Artifacts for contestability:** We define an artifact schema (decision logs, constraint reports, and policy manifests) that enables third-party audits, reconstruction of past decisions, and structured appeals.
- **Policy versioning as transparency:** We operationalize institutional value choices as a *versioned* policy (thresholds, guardrails, tie-breakers), enabling re-runs under new mandates without rewriting code.
- **Constraint-based multi-stakeholder governance:** We show how multiple ethical dimensions (fairness, operational capacity, transparency signals, and safeguards) can be negotiated as constraints rather than hidden in optimizer weights.
- **Socio-technical feasibility via trade-offs:** We provide a reproducible pipeline to surface the efficiency–fairness–oversight trade space, supporting deliberation rather than claiming a single "best" model.

2.2 Scope and claims

This is a *systems* contribution: we do not claim a new fairness definition or a new learning algorithm. Our claim is that contestability and accountability require *procedural artifacts* and *policy versioning* that are often missing from educational AI deployments. We evaluate the framework through controlled stress tests and an artifact-centric contestability scorecard, and we provide a reproducible fixed-score analyses, a tested reference application, and versioned artifact definitions.

The reported predictors estimate correctness or student success. They are mechanism testbeds, not validated rules for assigning support to the students at greatest risk. A prospective early-warning use requires a separately specified target, horizon and evaluation.

3 RELATED WORK

FAccT research has repeatedly shown that "getting the metric right" is insufficient when systems are embedded in institutions with asymmetric power, weak recourse, and opaque accountability. We position Compass-ECL-MASat the intersection of (i) contestability and procedural justice for algorithmic systems, (ii) principle-to-practice governance, and (iii) model-level fairness tooling and trade-offs.

3.1 Contestability, Due Process, and Institutional Accountability

A central FAccT concern is whether affected people can *understand, challenge, and obtain remedy* when algorithmic systems shape life chances. Citron and Pasquale argue that scoring systems can erode due process by obscuring decision logic and shifting power toward institutions [5]. Subsequent work emphasizes that accountability is not a property of an algorithm alone, but of the socio-technical system: what is documented, what is disclosed, who can audit, and what remedies exist [13, 17, 22]. In HCI and socio-technical design, contestability-by-design frameworks articulate practices that keep systems responsive to human intervention throughout the lifecycle [1].

Our contribution aligns with this line: we treat *audit artifacts* (policy manifests, constraint reports, decision traces) as first-class outputs required for contestation and institutional oversight.

3.2 From Principle-Level Ethics to Operational Governance

Major AI governance frameworks—NIST AI RMF, OECD AI Principles, UNESCO, and EU Trustworthy AI guidelines—converge on high-level desiderata (fairness, transparency, accountability, robustness) but remain implementation-agnostic [8, 15, 18, 19]. Scholars have highlighted the "last-mile" gap between ethical principles and operational mechanisms [6, 14]. We build on this insight, but focus on a specific public-sector domain where legitimacy and recourse are central: educational decision support.

3.3 Fairness Tooling, Trade-offs, and Preference-Based Selection

Model-level toolkits such as Fairlearn and Aequitas provide practical metrics, mitigation strategies, and reporting to support fairness evaluation [16, 21]. However, they do not by themselves specify institutional procedures for contestation, nor do they enforce policy versioning or create audit-ready decision artifacts. Trade-offs between utility, fairness, and other constraints are unavoidable; preference-based selection over Pareto-feasible candidates has been explored in multiple forms, including "fairness without harm" style decision rules and policy-oriented selection [20]. Our system-level contribution is to embed such choices into *versioned policy* and to couple selection with a procedural safeguard (abstention to human review) and an artifact trail.

3.4 Algorithmic Systems in Education

Educational prediction systems (early warning, risk scoring, intervention targeting) are widespread, but educational data often contains measurement and historical biases that complicate fairness claims [3]. Our setting therefore emphasizes governance practices

Table 2: Claims, evidence and explicit boundaries of the revised manuscript.

Claim	Evidence	Boundary
Fixed-score replay	State reconciliation and taxonomy replay, Tables 8 and 12	Does not support changed features or medical exceptions.
Empirical selection	Algorithm 1, Tables 9 and 10	Same-sample policy exploration; no population fairness guarantee.
Explicit failure	Revised runner gate and integration tests, Section 5	Historical violation logging was not operational fail-closed; no human queue connected.
Structured evidence benefit	Tables 5 and 6	Exploratory N=16, no formal ethics review; bundle effect, not isolated replay effect.
Entropy diagnostics	Table 13 and structured group-risk report	Scalar proxy; conditional row intervals, not cluster-robust field validation.

Table 3: Roles of related approaches; this conceptual positioning is not an exhaustive capability audit of all implementations.

Approach	Main role	Relationship to this work
Ethical principles	State institutional commitments	Motivate requirements, without fixing our schema.
Fairness toolkits	Measure or mitigate group disparities	Complement empirical ECL checks.
Impact assessments	Document impacts and responsibilities	Complement policy rationale and review procedures.
Contestability by design	Keep systems open to intervention	Motivates preserved case evidence and bounded replay.
COMPASS revision	Fixed-score policy audit and artifacts	Demonstrates explicit rules, reconstructed states and export gating.

(documentation, auditability, and recourse) in addition to model metrics.

Abstention, selective prediction, and distribution-free uncertainty. Our triage setting uses abstention ("reject option") to route uncertain cases to humans. This idea has a long history in statistical decision theory (e.g., Chow's rule) and has been revisited in modern *selective classification* and distribution-free uncertainty quantification (e.g., SelectiveNet; conformal prediction).[2, 4, 11] These lines of work motivate two design choices in our system: (i) treating high abstention as a *governance signal* rather than a silent knob, and (ii) recording the abstention policy in a replayable artifact so that institutions can audit when and why automation deferred to humans.

4 METHOD: POLICY, DECISIONS AND ARTIFACTS

4.1 Design scope

Our design makes institutional choices explicit, preserves evidence for audit, separates scoring from sensitive-attribute auditing, and treats abstention as a request for human consideration. The six COMPASS commitments—dignity, correctness, transparency, harmony, practical wisdom and active compassion—organize requirements; they are a domain-specific design vocabulary, not an exhaustive ethical theory. In the evaluated release the enforced numerical constraints are a fairness-gap cap and an automation-coverage floor. Documentation, intelligibility and institutional recourse remain separate requirements.

Table 4: Candidate decision states. A separate run-level gate determines whether export is authorized.

State	Accounting
`select`	Within capacity; counts toward selection and automation.
`reject`	Automated non-selection; counts toward automation.
`abstain`	Request for review; excluded from automation coverage.

4.2 Decision rule and units

A score s_i estimates $\Pr(y_i = 1 \mid x_i)$. All reported rankings are descending. Binary predictive entropy is

$$u_i = -q_i \log_2 q_i - (1 - q_i) \log_2 (1 - q_i), \tag{1}$$

where q_i clips s_i to $[10^{-7}, 1 - 10^{-7}]$ in float64. Entropy and γ are in bits; the maximum is one. Thus $\gamma = 0.999$ need not admit every case. Invalid probabilities and incompatible units are rejected.

The primary rule applies global abstention first: $u_i > \gamma$ yields `abstain`. Among eligible cases, the highest scores are selected up to capacity K. Equal scores are resolved by ascending persistent case ID. K is a ceiling: fewer than K cases may be selected. In allocation mode the remainder is `reject`; in screening mode it is `abstain`. The alternative Top-K-only abstention mode first nominates the global Top-K, abstains uncertain nominees and does not refill their places. These modes are explicit policy fields in the revised implementation.

Metrics. Let n_s, n_r, n_a count select, reject and abstain, with $n_s + n_r + n_a = n$. Automation coverage is $C = (n_s + n_r)/n = 1 - n_a/n$; selection rate is n_s/n. In screening mode $C \leq K/n$, so a floor $C_{\min} > K/n$ is structurally infeasible. Changing a deferral into a rejection changes institutional responsibility; it does not demonstrate greater educational benefit.

Algorithm 1 Revised empirical ECL–Pareto–MAS selection

Require: Candidates C, constraints, criteria T, tolerances δ_t
$F \leftarrow$ candidates with finite objectives satisfying constraints
if $F = \varnothing$ **then**
  return infeasible; close authorization gate
end if
$F \leftarrow$ nondominated candidates in F
for criterion t in T **do**
  $v^* \leftarrow$ best value of t in F
  $F \leftarrow \{c \in F : |t(c) - v^*| \le \delta_t\}$
end for
return candidate with smallest persistent ID in F

Utility@K is $\sum_{i:d_i=\text{select}} y_i/n_s$, undefined for empty selection. For binary audit groups we report the absolute gap between $\Pr(d = \text{select} \mid y = 1, a = 0)$ and $\Pr(d = \text{select} \mid y = 1, a = 1)$. Positive-label denominators include all cases in each group, not only eligible cases. A missing group or zero positive denominator produces insufficient evidence, not a zero gap. Labels support retrospective evaluation; they are not inputs required to replay a future decision.

4.3 ECL and the authorization boundary

ECL checks a candidate against declared empirical constraints. The paper sweep uses gap ≤ 0.05 and coverage ≥ 0.50. These are example normative choices, not universal safety thresholds. The historical synthetic recipe used a 0.60 coverage floor; the common replay analysis explicitly uses 0.50. Both reject its initial 0.004 coverage.

Historical runs recorded violations while retaining candidate decisions. The revised runner additionally writes an authorization gate: an infeasible or insufficient-evidence result cannot produce an authorized-decision export. Candidate packets remain available for diagnosis. A review-required record is generated, but no human queue is connected and no delivered review is claimed. Statistical feasibility on an audit sample does not certify future population fairness or legal compliance.

4.4 Candidate search and MAS

Candidates in the reported experiment reuse fixed scores and vary policy only. We distinguish the full 80-point (K, γ) grid from its 16-point fixed-K subset. Each candidate has an ID defined by dataset, capacity and threshold. The revised runner also supports an explicit policy grid and stores all candidates and the selection report.

Pareto dominance is exact over maximize utility, minimize absolute gap and minimize abstention. Equivalent candidates remain on the frontier. Fairness-first orders gap, utility, abstention; utility-first orders utility, gap, abstention. All reported tolerances are zero; the candidate ID is the final tie-breaker. These conventions replace historical underspecification and are recorded with each revised analysis. Repeated selection under an unchanged policy must be invariant to input row order. Changing criterion order tests policy sensitivity, not determinism by itself.

5 DESIGN PROPERTIES AND THEIR LIMITS

G1: Rule-aligned audit. Metrics and constraint checks use the decision states actually emitted by a specified rule. This prevents a training-score metric from being mistaken for deployed-rule compliance. Historical outputs and revised decisions are reconciled separately.

G2: Reproducible policy selection. With the same scored cases, IDs, candidate domain, objectives, policy and numerical conventions, the revised implementation applies exact Pareto filtering and deterministic tie-breaking. Tests include permuted cases and candidates. This is a computational property, not a claim of stability under new training samples or institutional preferences.

G3: Explicit failure. An empty feasible set or insufficient evidence closes the revised authorization gate. The historical code reported violations but did not establish that operational block. Neither version demonstrates a completed human review or institutional remedy.

G4: Evidence integrity. The revised verifier checks packet hashes, policy content, dataset-manifest bytes, fitted-model bytes, schema fields and decision replay. It never deserializes an untrusted model merely to verify its hash. Passing these checks establishes internal consistency, not truth of the source data, independent authenticity, absence of discriminatory proxies, or meaningful access to recourse.

All numerical constraints are retrospective sample checks. New cases require monitoring and an appropriate evaluation design; generalization guarantees are outside this paper. Raising an entropy threshold can increase automation while admitting more errors. Neither lower abstention nor successful replay establishes improved educational outcomes.

6 EXPERIMENTAL SETUP

6.1 Historical data and evaluation scope

We audit fixed scored test cases from three historical runs: ASSISTments interaction correctness ($n = 105107$), OULAD student success ($n = 6519$) and a synthetic binary stress test ($n = 1000$). ASSISTments uses $y = 1$ for a correct response and school-ID parity as a synthetic grouping proxy, not a protected attribute. OULAD uses $y = 1$ for Pass/Distinction and binary gender for auditing. Appendix 13 gives task and split details. These are outcome-prediction testbeds; descending success scores do not prioritize the highest-risk students for intervention.

The historical runner used a seeded shuffled 80/20 split and seed 42. We do not claim a temporal early-warning horizon, student-cluster independence for interaction rows, or prospective treatment effects. The revision replays emitted scores without retraining historical models. Policy search uses labels from the same retrospective audit sample, so selected operating points are exploratory, not an independently validated final policy.

6.2 Policy domains

All datasets use the threshold grid {.01, .05, .10, .15, .20, .30, .40, .50, .60, .70, .80, .90, .95, .99, .995, .999} bits. Capacity grids are ASSISTments $\{100, 250, 500, 1000, 2000\}$, OULAD $\{50, 100, 200, 500, 1000\}$ and synthetic $\{20, 50, 100, 200, 300\}$. Each full grid has 80 candidates. Fixed-budget recovery uses only the 16 thresholds at $K = 1000, 500, 100$, respectively. Full-grid profile comparisons are reported separately. We check gap $\le .05$ and automation coverage $\ge .50$ and apply the algorithm in Section 4.

6.3 Exploratory contestation study

The original study has 16 volunteers, eight standardized appeal scenarios and 64 case responses per condition. Profiles include engineering/data science, analytics and audit/compliance. Each participant responds to four cases under A (logs/cards) and four under B (structured per-case evidence, policy provenance and replay narratives). The author confirmed that G1 receives odd cases in A and even cases in B, with G2 reversed. Each case therefore has eight responses per condition. No participant sees the same case in both conditions. Detailed recruitment procedures, compensation and training records were not available in the inspected release.

For each case, participants choose a diagnostic label and remediation action, cite evidence, and rate confidence from 1 to 5. Evidence length is counted in Unicode characters, not words or a quality rubric. Time was self-reported for the questionnaire; we do not estimate per-case staff savings. The case materials include hypothetical exceptions and feature changes beyond the fixed-score replay engine. Consequently this study evaluates availability of structured evidence as a bundle, not execution of every scenario by the software or the isolated causal effect of replay.

Consent and ethics reporting. The author confirmed voluntary participation and authorization. The forms contain a consent item. No formal ethics-committee approval or exemption was obtained; we report this explicitly rather than infer an institutional authorization. No participant-level responses or free-text evidence are redistributed in the revised public analysis outputs. The setting and documentation limit the strength and generalizability of the study.

Retrospective analysis. We recompute participant-level condition averages from the original CSVs and case scoring keys. The revision uses two-sided signed-rank tests by enumerating all sign assignments to nonzero paired midranks; zero differences are excluded and equal absolute differences receive average ranks. No normal approximation or continuity correction is used. Holm adjustment covers diagnosis, remediation, confidence and evidence length. Paired mean differences have percentile 95% intervals from 10000 participant-bootstrap replicates (seed 42). This analysis was specified after inspecting historical results and is not preregistered. Diagnosis and remediation correctness coincide on all 128 case responses and are not independent replications. Software versions and exact counts are recorded with the released revision scripts.

7 RESULTS

We distinguish historical emitted decisions from the revised implementation and reanalysis. All numerical tables below are generated from the preserved packets or study responses; no historical predictor was retrained.

7.1 Structured evidence in the exploratory study

Table 5 reports recomputed averages, paired effects and revised statistical tests. Diagnostic and remediation accuracy increase from 0.5625 to 0.921875, a difference of 0.359375. Confidence increases from 3.078125 to 4.234375. These aggregates match the historical data; revised p-values differ because a single bilateral procedure and multiplicity correction replace the previously underspecified

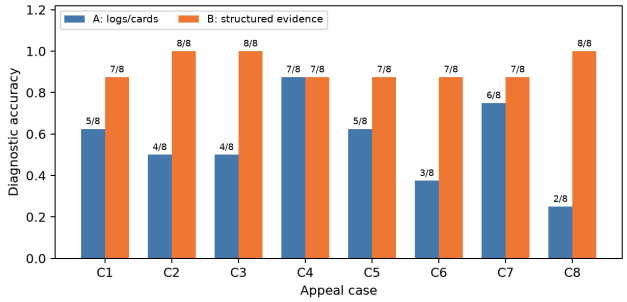


Figure 1: Diagnostic accuracy by case and condition. Labels give exact correct/total counts; all bars have eight responses.

tests. Longer evidence statements are descriptive and do not establish better reasoning. The experiment supports a benefit of the complete evidence bundle on these standardized tasks, with the limitations in Section 6.3.

7.2 Historical states and capacity

Table 7 recomputes the historical recipe outcomes and Table 8 gives exact counts. In the synthetic example $K = 100$ but only four cases are eligible and selected; coverage 0.004 is therefore consistent with the capacity ceiling. OULAD similarly selects 69 of a possible 500. Violations were recorded by the historical runner; those records alone do not prove operational escalation.

7.3 Fixed-budget recovery and variable-budget profiles

Table 9 separates full-grid and fixed-budget counts. Table 10 reports achieved metrics of the revised fixed-budget MAS choices. Raising γ recovers sample feasibility without retraining, demonstrating a policy-level route to resolve the particular violation. This does not exclude simultaneous model or measurement problems. Table 11 reports separate variable-budget comparisons with the complete profile definitions in Section 4. Full precision, group counts, exact grids and candidate IDs are in the structured revision report.

7.4 Taxonomy and uncertainty

Table 12 fixes scores, K and γ while changing abstention semantics. Utility stays fixed in this example, but automation coverage changes because non-selection can be an automated rejection or a deferral. Treating this as a policy correction requires institutional justification; the numbers alone cannot establish which responsibility assignment is appropriate.

Table 13 incorporates the previously disconnected numerical validation, recalculated with one entropy convention and average ranks for ties. OULAD has lower entropy/error monotonicity than ASSISTments; the synthetic predictor is poorly calibrated. Temperature scaling can worsen ECE, so we do not claim calibration robustness from preserved rank correlation. The report includes conditional row-bootstrap intervals and group-specific error counts

Table 5: Exploratory N=16 study; 64 case responses per condition. Participant means and paired differences with 95% participant-bootstrap intervals. p_H is the two-sided signed-rank sign-enumeration p-value adjusted by Holm over four endpoints; diagnosis/remediation correctness coincides in all 128 responses.

Outcome	A	B	Difference [95% CI]	p_H
Diagnosis	0.562	0.922	0.359 [0.203, 0.516]	0.00293
Remediation	0.562	0.922	0.359 [0.203, 0.516]	0.00293
Confidence	3.078	4.234	1.156 [0.922, 1.375]	0.00012
Evidence (characters)	69.812	91.656	21.844 [20.047, 23.734]	0.00012

Table 6: Diagnostic counts by case and condition. Every cell has eight responses; case materials are standardized research scenarios.

Case	A	B
C1	5/8	7/8
C2	4/8	8/8
C3	4/8	8/8
C4	7/8	7/8
C5	5/8	7/8
C6	3/8	7/8
C7	6/8	7/8
C8	2/8	8/8

and Wilson intervals across thresholds. Historical interaction packets lack student cluster IDs, so the ASSISTments intervals do not establish cluster-robust uncertainty or field reliability.

7.5 Artifact checks and appeal scope

The historical four-point scorecard was populated by constant booleans in the runner, so it is not evidence of independent artifact validation. The revised verifier instead computes integrity and decision-reconstruction checks and tests failures after packet tampering or missing policy content. Fitted-model hashes are separated from configuration hashes; the earlier configuration-only digest is not presented as an immutable trained-model identity.

The appeal pack (Table 14) is a set of study materials. Medical exceptions, changing income, or removing a feature require additional policy inputs or re-scoring and are outside the fixed-score replay contract. Evidence strings in these materials are not execution logs proving that the released engine implements those operations.

8 DISCUSSION AND LIMITATIONS

8.1 Institutional value and responsibility

An audit trail makes choices inspectable, but does not make the choices legitimate by itself. Constraints are set by actors with authority and incentives; versioning can expose changes but cannot prevent capture or selective compliance [13, 17]. Institutions still need accessible appeal channels, a mandate for policy changes, appropriate staffing and a way to reverse or remedy decisions. A deployment should record who changed a policy, why, who may contest it and how resulting cases are handled.

Automation coverage measures automated outcomes, including rejection. It must not be equated with access to support or students helped. In screening mode the coverage floor can conflict structurally with limited K; increasing automation by relabeling cases changes the division of responsibility between software and staff. Our experiments expose that choice rather than endorse one taxonomy universally.

8.2 What the evidence establishes

The strongest evidence is computational: rules can be made explicit, decisions can be replayed on fixed scores, infeasibility can be surfaced, and revised exports can be gated. Sample-level constraints are not population guarantees. Policy selection on the same labelled audit set is exploratory. One historical model and split per dataset do not establish multi-seed predictive stability, comparative state of the art or educational treatment benefit.

The N=16 study is exploratory, uses author-defined standardized scenarios and mixed technical profiles, and has incomplete recruitment and training documentation. Volunteers and consent do not substitute for a formal ethics review, which was not obtained. No causal separation among provenance, structured information and replay is possible in its two conditions. Diagnosis and remediation correctness are perfectly concordant in these responses, so the study provides fewer independent outcome dimensions than its labels might suggest. A prospective study with independent cases, frontline stakeholders and documented institutional review is needed before making field claims.

8.3 Uncertainty and data limitations

Predictive entropy is a scalar confidence proxy, not a decomposition of epistemic and aleatoric uncertainty. Calibration and entropy/error correlation do not alone validate the safety of deferral. We therefore provide group-level selective-error summaries and explicitly conditional intervals. ASSISTments uses school-ID parity only to stress the constraint machinery; it does not measure protected-group disparities. OULAD uses binary gender and success labels, leaving wider intersectional and measurement questions unresolved [3].

8.4 Privacy, integrity and operational boundaries

Packets must contain the inputs required for the supported rule, with purpose-appropriate access and retention controls. Publishing medical information or raw feature histories is not required by our minimum contract. An institution can keep restricted source

Table 7: Historical recipe outcomes, recomputed from emitted states. Entropy thresholds are in bits; VIOL indicates a retrospective ECL violation, not a delivered human review.

Dataset	K	γ	Utility	Coverage	Abstain	Gap	ECL
ASSISTments	1000	0.60	0.981	0.758560	0.241440	0.00497	OK
OULAD	500	0.60	0.174	0.010584	0.989416	0.00192	VIOL
Synthetic	100	0.65	1.000	0.004000	0.996000	0.00895	VIOL

Table 8: Exact initial state counts. K is a capacity ceiling, not the realized number selected.

Dataset	n	Eligible	Select	Reject	Abstain
ASSISTments	105107	79730	1000	78730	25377
OULAD	6519	69	69	0	6450
Synthetic	1000	4	4	0	996

Table 9: Candidate counts under gap ≤ 0.05 and coverage ≥ 0.50. Full grids vary K and γ; fixed-budget analyses use only the 16 thresholds at the recipe budget.

Dataset	Full	Feas.	Fixed	Feas.	Pareto
ASSISTments	80	55	16	11	1
OULAD	80	15	16	3	1
Synthetic	80	12	16	3	1

evidence behind resolvable references, but missing evidence must be reported rather than treated as replay success. Hashes establish consistency of bytes, not external authenticity or truth. The revised runner's review-required file is not a connected case-management process.

We make no automatic legal-compliance claim. A particular deployment requires jurisdiction-specific institutional review. Instrumented per-case audit cost, staffing capacity, actual access to appeal, independent prospective evaluation and model re-scoring under corrected inputs remain future work. Changing a feature or target creates a new inference or modeling task and must not be described as a policy-only replay.

9 CONCLUSION

We present an artifact-oriented approach to educational decision-system governance: explicit policy, empirical constraints, deterministic feasible-Pareto selection and fixed-score replay. The revision reconciles definitions, identifies the actual outcome-prediction tasks, specifies entropy in bits and capacity ceilings, and separates historical violation reporting from the revised export gate. Computational reports expose policy-dependent feasibility and the consequences of decision taxonomy without implying improved student outcomes.

An exploratory N=16 study supports the complete structured-evidence bundle for standardized appeal diagnosis, with revised statistical reporting and explicit ethical and methodological limits. It does not validate every hypothetical exception, isolate replay from information availability or establish institutional recourse. Future work requires prospective stakeholder evaluation, documented ethical review, independent policy validation, richer uncertainty methods and an instrumented human workflow. Code, exact counts and revision analyses are identified in Appendix 10; distribution of participant responses is separate from distribution of aggregate results.

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Table 10: Revised MAS selections from the fixed-K feasible Pareto set. All tie tolerances are zero; persistent candidate ID is the final tie-breaker.

Dataset	K	γ	Selected	Utility	Gap	Coverage	Abstain
ASSISTments	1000	0.999	1000	0.981	0.00497	0.999448	0.000552
OULAD	500	0.999	500	0.732	0.01345	0.906427	0.093573
Synthetic	100	0.999	100	1.000	0.01697	0.933000	0.067000

Table 11: Variable-K profiles over the full 80-point grids; these are distinct from fixed-budget recovery. Fairness-first orders gap, utility, abstention; utility-first orders utility, gap, abstention.

Dataset	Profile	K	γ	Utility	Gap	Coverage
ASSISTments	Fairness-first	100	0.999	0.980	0.00040	0.999448
ASSISTments	Utility-first	250	0.999	0.992	0.00206	0.999448
OULAD	Fairness-first	50	0.999	0.800	0.00685	0.906427
OULAD	Utility-first	100	0.999	0.830	0.00754	0.906427
Synthetic	Fairness-first	20	0.999	1.000	0.00020	0.933000
Synthetic	Utility-first	20	0.999	1.000	0.00020	0.933000

Table 12: ASSISTments taxonomy replay with fixed scores, $K = 1000$ and $\gamma = 0.60$ bits. Changes in automation coverage represent changes in decision responsibility, not demonstrated improvement in student outcomes.

Policy	Utility	Coverage	Abstain	Gap
Global abstention	0.981	0.758560	0.241440	0.00497
Top-K-only abstention	0.981	1.000000	0.000000	0.00497
Defer remainder	0.981	0.009514	0.990486	0.00497

Table 13: Recomputed entropy validation. ECE uses 10 equal-width probability bins including score 1; Spearman uses average ranks for ties. Both baseline and temperature stress use entropy in bits.

Dataset	ECE	ρ	ECE ($T = 2$)	ρ ($T = 2$)
ASSISTments	0.0271	0.9789	0.0905	0.9818
OULAD	0.0128	0.4166	0.0703	0.4166
Synthetic	0.2914	0.6774	0.2398	0.6774

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10 REPRODUCIBILITY AND REVISION EVIDENCE

The revision preserves the historical sources and numerical assets in a checksum-indexed snapshot. A version inventory and work log are maintained in the accompanying `revisao_artigo` directory. The reference repository is https://github.com/rtales/compass-contestable-governance-artifacts; local revised code and hashes, rather than the unmodified remote HEAD, identify the changes evaluated here.

The historical packet runs used are ASSISTments, OULAD and synthetic snapshots from 26 January 2026, all seed 42. Their full

Table 14: Illustrative N=16 study materials, not execution logs. Full Portuguese narratives and A/B evidence strings remain in the eight local case JSONs. The scope column distinguishes fixed-score replay from operations that the engine does not implement.

Case	Diagnostic category	Evidence illustrated in the instrument	Engine scope
C1	Stale input data	Income snapshot and update lineage	Corrected inputs require new inference.
C2	Policy change	Policy versions and effective dates	Fixed-score replay only for supported policy fields.
C3	Normative exception	Medical flag and force-inclusion rule	Exception rule and medical inputs are not implemented.
C4	Capacity constraint	School capacity and revised availability	Global K replay supported; per-school capacity is outside the engine.
C5	Fairness constraint	Group gap and policy cap	Retrospective candidate check; no per-person fairness reranking.
C6	Proxy risk	ZIP-code correlation and feature removal	Proxy audit/removal requires additional analysis or fitting.
C7	Abstention error	Uncertainty value and threshold	Supported score-based abstention reconstruction.
C8	Version mismatch	Declared and executed policy hashes	Content/hash consistency check; no CI/CD system integration.

paths and SHA-256 digests appear in the computational revision report. No new fit of those historical models is claimed.

Rebuilding the analysis. From the project root, the script `revisao_artigo/scripts/recalcular_evidencias.py` regenerates computational reports, candidate tables, group-risk summaries, study aggregates and the English diagnostic figure. It uses the preserved case packets and, for the study, the author's local response ZIP. The author confirmed the original assignment mapping in this revision. Outputs are in `paper_assets/revised/`. Individual response text is not copied to those outputs.

The report records complete thresholds, capacities, IDs, counts and comparisons with historical CSVs. Fixed-budget and full-grid results have separate domains. Script source and an environment lock accompany the revision. Application tests include a synthetic end-to-end run and corruption checks; they do not substitute for re-fitting real-data predictors.

Access limitations. Historical packet labels and audit groups are retrospective evaluation data. Participant responses remain local in this revision; sharing consent and institutional rules must be checked before redistribution. Aggregate counts and scripts are included. A computational reproduction can run without the study data and must report the study as skipped, never silently substitute generated participants. The revised study analysis was retrospective, not preregistered.

Compilation and verification. The multi-file LaTeX project is compiled with Tectonic, with its release recorded in the revision tools inventory. Generated tables use unrounded metrics and round only for display. All rendered pages are inspected for clipping and broken references. Source snapshots, checksums, software versions and verification records distinguish the historical release from this corrected draft.

11 IMPLEMENTATION AND VERSION-SPECIFIC INVENTORY

The historical release contains one scored run per dataset, metrics, metadata, decision briefs and case packets. Sweeps and illustrative appeal materials exist in the paper project, and original N=16 responses are available locally to the author. The previous statement that the release lacked sweeps and human evaluation was inconsistent with these materials and has been removed. The study AIES N=18 is a different version and is not used here.

The revised application is in the repository directory named `compass-contestable-governance-artifacts`. New runs persist policy content, a dataset-manifest copy, fitted-model bytes, separate configuration/model hashes, source-file hashes, selection reports, candidate packets, computed artifact checks and an authorization gate. A Git HEAD alone does not identify uncommitted source changes. Model snapshots are hashed as opaque bytes; verifying integrity does not require loading untrusted serialized code.

A feasible run may export authorized candidate decisions. Infeasible or insufficient-evidence runs preserve evaluation evidence and write a review-required record without an authorized export. They do not claim delivery to a human queue. Candidate metrics remain distinguishable from the gate result.

No broad predictive-baseline study, multi-seed model evaluation, field deployment or individually timed appeal-cost study is claimed. Historical outputs are preserved; revision reports identify which values match and which procedures changed.

12 PROSPECTIVE EARLY-WARNING BLUEPRINT: NOT EVALUATED HERE

A future dropout-prevention use would define a risk-positive outcome, such as Fail/Withdrawn, an intervention horizon, features available before that horizon, and ranking aligned to that objective. It would also require temporal and student-aware splits, label audits and institutionally justified sensitive attributes. These choices

differ from the implemented OULAD success target and descending success-score ranking. The current utility values must not be reused as evidence for this proposed task.

Likewise, interaction correctness in ASSISTments is not equivalent to a per-student need-for-support label. Aggregating interactions, selecting an eligible population, specifying the treatment and measuring actual benefit require a separate evaluation. Changing the target or removing features can require retraining; changing an input value requires new inference. Neither operation is a fixed-score policy replay.

13 DATASETS AND IMPLEMENTED TASKS

13.1 ASSISTments

The 2009–2010 Skill Builder release provides student–problem interactions [10]. The implemented target is response correctness (positive=1). School-ID parity is a synthetic group for testing constraints, not a demographic or protected attribute.

The processor reads the CSV with a deterministic encoding fallback, replaces decoding errors and skips malformed lines. Numeric missing values are filled with zero, including skill and response-time fields; the implementation does not drop rows for those missing fields. Correctness is coerced to an integer and clipped to zero/one; missing labels become zero. These choices can affect measurement validity and should be audited before deployment.

Features include bounded attempt and hint counts, log-transformed response and overlap times, practice opportunity, an original-item indicator, skill-ID remainder, and fixed one-hot categories for tutor mode, answer type and task type. School ID itself is excluded from the feature vector, but that does not establish absence of proxies.

The runner applies a seed-42 shuffled row split (80/20); the audited output has 105107 cases. Features can describe the interaction being predicted. This is not a temporally validated early-warning task. The historical packets lack student-cluster IDs, so neither split independence nor cluster-robust intervals are established.

13.2 OULAD

OULAD provides anonymized student/course information [12]. The processor uses the student-information table. Positive means Pass or Distinction; Fail and Withdrawn are negative. Gender F/M defines the binary audit group, with other or missing gender excluded.

Categorical inputs are module, presentation, region, education, deprivation band, age band and disability; numerical inputs are previous attempts and studied credits. Gender and final outcome are excluded from the score features. This is a success-prediction benchmark; it does not use a validated early-intervention time window or estimate treatment benefit. The shuffled 80/20 row split yields 6519 audited cases. Repeated student/course records are not claimed to be separated by student.

13.3 Synthetic

The synthetic audit has 1000 held-out cases and a binary generated group. Its purpose is to exercise capacity, abstention, infeasibility and evidence checks under a controlled label process. It is not evidence of field validity. Recipe inputs, hashes and exact group counts are included in the computational revision report.

14 ARTIFACT CONTRACT AND REPLAY PROTOCOL

14.1 Minimum fixed-score contract

Schema 2.0 records a persistent case ID, finite score, entropy in bits, clipping convention, decision state, supported replay scope, policy reference/hash, dataset-manifest hash, fitted-model hash and packet hash. Policy content records capacity, score direction, allocation/screening mode, global/Top-K-only abstention, threshold/unit and selection criteria/tolerances. Hashes alone are not substitutes for accessible policy content. Labels and audit groups are evaluation fields, separate from decision inputs.

Historical packets have a smaller unversioned schema. Their entropy is checked against the known bit-based implementation before reanalysis; a legacy configuration digest is not described as a fitted-model digest. The revision does not retroactively claim that old packets implemented schema 2.0.

14.2 Reconstruction and validation

Replay reads the same fixed scores and case IDs, applies the declared rule and compares every emitted state with the original. The revised verifier additionally checks content hashes, schema/unit fields, policy availability, manifest bytes and model-snapshot bytes. Missing, altered or incompatible evidence causes failure, not a perfect score by default. Tests include score ties, reordered inputs, boundary thresholds, empty feasible sets and corrupted packets.

14.3 Scope of appeal materials

Medical-force-inclusion rules require the applicable rule, its version, the relevant flag and exception precedence. These are absent from the minimum score-based contract; no executed medical-exception replay is claimed. Likewise, income corrections or feature removal require updated inputs and re-scoring, possibly retraining. The study's evidence strings are scenario materials rather than logs proving these operations were executed. Restricted evidence may remain outside public bundles, but its availability and access conditions must be documented.

14.4 Uncertainty analysis

ECE uses ten equal-width probability bins, including probability one in the final bin. Spearman entropy/absolute-error correlation uses average ranks for ties. Temperature stress uses logits scaled by $T \in \{0.5, 2\}$ and recomputes entropy in bits. Conditional percentile intervals use 200 row-bootstrap replicates, seed 42. Per-group selective risk is the binary classification error among entropy-eligible cases, not precision among Top-K selections; reports give error counts, denominators and descriptive Wilson intervals. Interaction packets do not preserve student cluster IDs, so these row-based intervals do not establish cluster-robust population uncertainty.